STREAKIT — BUILDING ROADMAP

Phase 2: Actual Build

Risk & Solution Reference

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1. PROJECT OBJECTIVE

StreakIt is an interactive virtual microbiology lab designed to help microbiology students practice experiments repeatedly through interactive simulations, AI-assisted learning, personalized feedback, progress tracking, and streak-based engagement.

Core Product Loop

Learn → Perform → Make mistakes → Get feedback → Improve → Progress → Return

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2. BUILD ROADMAP

STEP 01 — Project Setup

- Create the project.
- Set up the simplest reliable tech stack.
- Confirm the website launches successfully.

STEP 02 — Product Structure

Build the basic:

- Landing page
- Onboarding
- Dashboard
- Experiment library
- Navigation

STEP 03 — Build ONE Experiment First

- Create the virtual lab environment.
- Add relevant equipment.
- Make objects interactive.
- Track student actions.
- Validate correct/incorrect actions.

Do not build all five experiments yet.

STEP 04 — Learning & Test Modes

Learning Mode

- AI assistance available.
- Hints and explanations allowed.

Test Mode

- No AI assistance.
- No hints.
- Student performs independently.
- System evaluates the result afterward.

STEP 05 — AI Learning Assistant

- Answer microbiology questions.
- Explain concepts simply.
- Explain why a step matters.
- Help students understand mistakes.
- Do not perform the experiment for them.

STEP 06 — Personalized Feedback

After an experiment:

- Accuracy
- Successful steps
- Mistakes
- Struggle areas
- Time taken
- Recommended practice

STEP 07 — Streak & Progress

- Experiment completion
- Current streak

- Progress dashboard
- One simple daily activity/challenge
- Reason to return tomorrow

STEP 08 — Remaining Experiments

Once the first experiment works properly, adapt the same system for the remaining four.

STEP 09 — Integration & Testing

Test:

- Navigation
- Experiments
- AI
- Learning/Test modes
- Feedback
- Progress
- Streaks
- Data saving

STEP 10 — Presentation & Demo

Prepare:

- Live demo
- Backup recorded demo
- Screenshots
- Pitch deck
- Presentation
- Product demo/ad

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3. CORE FIVE EXPERIMENTS

1. **Gram Staining**
2. **Streak Plate Method**
3. **Antimicrobial Susceptibility Test**
4. **Serial Dilution & Pour Plate**
5. **Biochemical Testing for Species Identification**

Note: Final definitions and exact scope of each experiment must be verified through Javeria's research and the microbiology-student tester.

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4. RISKS

AI / Coding

- AI generates broken code.
- AI misunderstands requirements.
- AI modifies working features unnecessarily.
- AI creates visually convincing but fake functionality.
- AI coding tools struggle with large/complex projects.

Product

- Experiment becomes too complicated.
- Too many features are added.
- Design takes too long.
- Features don't integrate properly.

Scientific

- AI gives incorrect microbiology information.
- Experiment procedure is scientifically inaccurate.
- Virtual interaction doesn't represent the real procedure correctly.

Technical

- Dependencies conflict.
- 404/routing errors occur.
- Frontend and backend don't communicate.
- Data doesn't save correctly.
- AI/API becomes unavailable or slow.
- Website becomes slow.
- Urdu/RTL layout breaks.
- Responsive layout breaks.

Demo

- Live website fails.
- Internet/API unavailable.

- Data or functionality breaks during presentation.

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5. SOLUTIONS

Risk	Solution
Broken AI code	Build one feature at a time and test immediately.
AI misunderstands requirements	Give exact behavior, inputs, outputs and examples.
AI changes working features	Tell it to modify only relevant files/features and keep versions/backups.
Fake functionality	Test actual behavior, not just appearance.
AI tool struggles	Break the project into small modules and focused prompts.
Too much complexity	Freeze the MVP and cut unnecessary features.
Feature creep	Put extra ideas into a **Later** list.
Design takes too long	Design MVP screens first; polish later.
Scientific inaccuracies	Use verified research and have the microbiology-student tester check experiments.
Wrong AI answers	Give the AI verified StreakIt research/context rather than letting it freely invent answers.
AI helps in Test Mode	Explicitly enforce separate Learning Mode and Test Mode behavior.
Dependency problems	Use the simplest stack and avoid unnecessary packages.
404 error	Check the route, page, URL and deployment configuration.
Frontend/backend mismatch	Test the connection early with simple data.
Data doesn't save	Test save/load before building additional features.
AI/API failure	Provide a fallback state; don't make the entire product dependent on one API call.

Slow website	Optimize images, animations, API calls and unnecessary libraries.
Urdu/RTL problems	Keep Urdu scope controlled and test RTL early.
Integration problems	Connect features continuously instead of waiting until the end.
Demo failure	Keep live demo + backup video + screenshots.

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6. NON-NEGOTIABLE BUILD RULES

RULE 1

Build → Test → Verify → Save → Move on.

RULE 2

Never ask the AI tool to **build the entire product at once.**

RULE 3

Don't change multiple major systems simultaneously.

RULE 4

Keep verified scientific information separate from AI assumptions.

RULE 5

Learning Mode = assistance.

Test Mode = independent performance.

RULE 6

If a feature doesn't help prove the StreakIt concept, **cut it from the MVP.**

RULE 7

Always keep a working version before making a major change.

RULE 8

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Never leave demo preparation until the final day.

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7. MVP DEFINITION

MVP = Minimum Viable Product.

For StreakIt, the MVP is the smallest usable version that proves:

> **Students can practice microbiology experiments virtually, receive AI-assisted learning, receive personalized feedback, track progress/streaks, and have a reason to return.**

It does **not** need every future feature.

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8. STREAKIT'S CORE EXPERIENCE

Landing

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Personalized Onboarding

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Dashboard

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Experiment Library

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Experiment Brief

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Learning Mode / Test Mode

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🧪 **Interactive Virtual Experiment**

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🤖 **AI Learning Assistant**

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📊 **Personalized Feedback**

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🔥 **Streak + Progress**

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Daily Challenge / Next Activity

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Return Tomorrow

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9. FINAL BUILD MINDSET

We are **not building the final StreakIt.**

We are building the **first working proof that StreakIt works.**

The goal is not:

> *"Everything must be perfect."*

The goal is:

> **"The core experience must actually work."**

Keep it simple. Build it. Test it. Fix it. Ship it. 🧪🔥